VU Quiz Collection

Total Quizzes: 10

Courses: MTH401 - Differential Equations (Quiz 2), MTH603 - Numerical Analysis (Quiz No 02)

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Quiz 1 - MTH401 - Differential Equations (Quiz 2)

Question:

If y=2+xy=2+xy = 2 + x , then which of the following is true for it?

Options:

A. Its annihilator operator is D+1

B. Its annihilator operator is D3D3{D^3}

C. Its annihilator operator is D

D. Its annihilator operator is D2D2{D^2}

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Quiz 2 - MTH401 - Differential Equations (Quiz 2)

Question:

If y=c1ex+c2e4xy=c1ex+c2e4xy = {c_1}{e^x} + {c_2}{e^{4x}} is the complementary solution of y 2 2" 5 y 2 + 4 y = 8 e x y 3" 5 y 2 + 4 y = 8 e x y'' - 5y' + 4y = 8{e^x} , then the general form o ypyp{y_p}=_____.

Options:

A. None of these

B. AxexAxexAx{e^x}

C. AexAexA{e^x}

D. (Ax+B)e4x(Ax+B)e4x(Ax + B){e^{4x}}

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Quiz 3 - MTH401 - Differential Equations (Quiz 2)

Question:

Considerthenon - homogeneouslineardifferentialequation,any(n)+an" 1y(n" 1)+.

g(x)=5x+7,thentheassumedparticularsolutionypcouldbe_____.Considerthenon - homogeneouslineardifferential equation,any(n)+an" 1y(n" 1)+...+a1y2+a0y=g(x). Iftheinputfunction,g(x)=5x+7,thentheassumedouldbe_____.\begin{gathered}
{\text{Consider}}\,\,{\text{the}}\,\,{\text{non - homogeneous}}\,\,{\text{linear}}\,\,{\text{differential}}\,\,{\text{equation,}}\,\,{a_n}{y^{(n)}} + {a_{n - 1}}{y^{(n - 1)}} + ... + {a_1}y' + {a_0}y = g(x).\,\, \hfill \\\ If\,\,{\text{the}}\,\,{\text{input}}\,\,{\text{function,}}\,\,g(x) = 5x + 7,\,\,{\text{then}}\,\, {\text{the}}\,\,{\text{assumed}}\,\,{\text{particular}}\,\,{\text{solution}}\,\,{y_p}\,\,{\text{could}}\,\,{\text{be}}\,\,{\text{_____}}{\text{.}}\, \hfill \\\\
\end{gathered}

Options:

A. AAA

B. Ax2+Bx+CAx2+Bx+CA{x^2} + Bx + C

C. Ax3+Bx2+Cx+DAx3+Bx2+Cx+DA{x^3} + B{x^2} + Cx + D

D. Ax+BAx+BAx + B

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Quiz 4 - MTH401 - Differential Equations (Quiz 2)

Question:

Considerthenon - homogeneouslineardifferentialequation,any(n)+an" 1y(n" 1)+.g(x)=3x2" 2,thentheassumedparticularsolutionypcouldbe_____.Considerthequation,any(n)+an" 1y(n" 1)+...+a1y2+a0y=g(x). Iftheinputfunction,g(x)=3x2" 2couldbe_____.\begin{gathered}
{\text{Consider}}\,\,{\text{the}}\,\,{\text{non - homogeneous}}\,\,{\text{linear}}\,\,{\text{differential}}\,\,{\text{equation,}}\,\,{a_n}{y^{(n)}} + {a_{n - 1}}{y^{(n - 1)}} + ... + {a_1}y' + {a_0}y = g(x).\,\, \hfill \\\ If\,\,{\text{the}}\,\,{\text{input}}\,\,{\text{function,}}\,\,g(x) = 3{x^2} - 2,\,\,{\text{then}}\,\,{\text{the}}\,\,{\text{assumed}}\,\,{\text{particular}}\,\,{\text{solution}}\,\,{y_p}\,\,{\text{could}}\,\,{\text{be}}\,\,{\text{_____}}{\text{.}}\, \hfill \\\\
\end{gathered}

Options:

A. AAA

B. Ax2+Bx+CAx2+Bx+CA{x^2} + Bx + C

C. Ax+BAx+BAx + B

D. Ax3+Bx2+Cx+DAx3+Bx2+Cx+DA{x^3} + B{x^2} + Cx + D

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Quiz 5 - MTH401 - Differential Equations (Quiz 2)

Question:

In the population growth model, the solution of the differential equation is given by _____.

Options:

A. A(t)=A0/2A(t)=A0/2A(t) = {A_0}/2

B. A(t)=A0ektA(t)=A0ektA(t) = {A_0}{e^{kt}}

C. A(t)=kektA(t)=kektA(t) = k{e^{kt}}

D. A(t)=A0/2A(t)=A0/2A(t) = {A_0}/2

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Quiz 6 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

While using Jacobi method for the matrix
A =#¡#£#¢ 1 1 / 4 1 / 2 1 / 4 1 / 3 1 / 4 1 / 2 1 / 4 1 / 5#¤#¦#¥ A = [1 1 / 4 1 / 2 1 / 4 1 / 3 1 / 4 1 / 2 1 / 4 1 / 5] A = \left[1&{1/4}&{1/2}\\{1/4}&{1/3}&{1/4}\\{1/2}&{1/4}&{1/5}
\end{array}} \right]
and ‘theta ¸ =0.4480, the orthogonal matrix S1 will be given by

Options:

A. S1=#¡#£#¢cos0.44800sin0.44800 10" sin0.44800cos0.4480#¤#¦#¥ S1=[cosa0.4480sa0.4480]{S_1} = \left[{\begin{array}{*{20}{c}}
{\cos 0.4480}&0&{ - \sin 0.4480}\\0&1&0\\{\sin 0.4480}&0&{\cos 0.4480}
\end{array}} \right]

B. S1=#¡#£#¢cos0.4480sin0.44800" sin0.4480cos0.44800001#¤#¦#¥ S1=[cosa0.4480001]{S_1} = \left[{\begin{array}{*{20}{c}}
{\cos 0.4480}&{ - \sin 0.4480}&0\\{\sin 0.4480}&{\cos 0.4480}&0\\0&0&1
\end{array}} \right]

C. S1=#¡#£#¢1000cos0.4480sin0.44800" sin0.4480cos0.4480#¤#¦#¥ S1=[1000cosa0.4480sa0.4480]{S_1} = \left[{\begin{array}{*{20}{c}}
1&0&0\\0&{\cos 0.4480}&{ - \sin 0.4480}\\0&{\sin 0.4480}&{\cos 0.4480}
\end{array}} \right]

D. S1=#¡#£#¢cos0.44800000" sin0.448000001#¤#¦#¥ S1=[cosa0.44800000" sina0.44800
{\begin{array}{*{20}{c}}

{\cos 0.4480}&0&0\\0&{ - \sin 0.4480}&0\\0&0&1
\end{array}} \right]

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Quiz 7 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

While using Jacobi method for the matrix
A =#¡#£#¢ 2" 1 0" 1 2 0 0 0 2#¤#¦#¥ A = [2" 1 0" 1 2 0 0 0 2] A = \left[{\begin{array}{*{20}{c}}
2&{ - 1}&0\\{ - 1}&2&0\\0&0&2
\end{array}} \right]
the value of 'theta ¸' can be found as

Options:

A. tan2¸=2a13a11" a33tana2¸=2a13a11" a33\tan 2\theta = \frac{{2{a_{13}}}}{

B. tan2¸=2a23a22" a33tana2¸=2a23a22" a33\tan 2\theta = \frac{{2{a_{23}}}}{

C. tan2¸=2a12a11" a22tana2¸=2a12a11" a22\tan 2\theta = \frac{{2{a_{12}}}}{

D. tan2¸=2a13a33" a11tana2¸=2a13a33" a11\tan 2\theta = \frac{{2{a_{13}}}}{

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Quiz 8 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

By using determinants, we can easily check that the solution of the given system of linear equation exits and it is unique.

Options:

A. FALSE

B. TRUE

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Quiz 9 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

Which of the following systems of linear equations has a strictly diagonally dominant coefficient matrix?

Options:

A . " 5 x 1 + 3 x 2 + 2 x 3 = 1 7 x 1 + 5 x 2" 8 x 3 = 9 4 x 1 + 2 x 2 + 3 x 3 = 1" 5 x 1 + 3 x 2 + 2 x 3 = 1 7 x 1 + 5 x 2" 8 x
array}{l}
- 5{x_1} + 3{x_2} + 2{x_3} = 1\\7{x_1} + 5{x_2} - 8{x_3} = 9\\4{x_1} + 2{x_2} + 3{x_3} = 1
\end{array}

B . 4 x 1 + 2 x 2 + 3 x 3 = 1" 5 x 1 + 3 x 2 + 2 x 3 = 1 7 x 1 + 5 x 2" 8 x 3 = 9 4 x 1 + 2 x 2 + 3 x 3 = 1" 5 x 1 + 3 x 2 + 2
array}{l}
4{x_1} + 2{x_2} + 3{x_3} = 1\\ - 5{x_1} + 3{x_2} + 2{x_3} = 1\\7{x_1} + 5{x_2} - 8{x_3} = 9
\end{array}

C . 7 x 1 + 5 x 2" 1 x 3 = 9" x 1 + 3 x 2 + x 3 = 1 x 1 + 2 x 2 + 4 x 3 = 1 7 x 1 + 5 x 2" 1 x 3 = 9" x 1 + 3 x 2 + x 3 = 1 x 1
7{x_1} + 5{x_2} - 1{x_3} = 9\\ - {x_1} + 3{x_2} + {x_3} = 1\\{x_1} + 2{x_2} + 4{x_3} = 1
\end{array}

D . 4 x 1 + 2 x 2 + 3 x 3 = 1 7 x 1 + 5 x 2" 8 x 3 = 9" 5 x 1 + 3 x 2 + 2 x 3 = 1 4 x 1 + 2 x 2 + 3 x 3 = 1 7 x 1 + 5 x 2" 8 x
array}{l}
4{x_1} + 2{x_2} + 3{x_3} = 1\\7{x_1} + 5{x_2} - 8{x_3} = 9\\ - 5{x_1} + 3{x_2} + 2{x_3} = 1
\end{array}

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Quiz 10 - MTH603 - Numerical Analysis (Quiz No 02)

Question:

While using Jacobi method for the matrix
A =#¡#£#¢ 2 0 0 0 2 1 0 1 2#¤#¦#¥ A = [2 0 0 0 2 1 0 1 2] A = \left[{\begin{array}{*{20}{c}}
2&0&0\\0&2&1\\0&1&2
\end{array}} \right]
the value of ‘theta ¸’ can be found as

Options:

A. tan2¸=2a13a11" a33tana2¸=2a13a11" a33\tan 2\theta = \frac{{2{a_{13}}}}{

B. tan2¸=2a12a11" a22tana2¸=2a12a11" a22\tan 2\theta = \frac{{2{a_{12}}}}{

C. tan2¸=2a23a22" a33tana2¸=2a23a22" a33\tan 2\theta = \frac{{2{a_{23}}}}{

D. tan2¸=2a13a33" a11tana2¸=2a13a33" a11\tan 2\theta = \frac{{2{a_{13}}}}{{{a_{33}} - {a_{1

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